SAI ARUN GUNUGANTI

Lead Software Engineer | Frontend Platform, Micro-Frontends & Agentic Systems

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SUMMARY

Lead Software Engineer with 10+ years building large-scale web platforms for retail, banking, and healthcare (Lowe's, Apple, Cisco, McKesson, M&T Bank). Currently Lead Developer (team of 5) at Lowe's, owning a checkout-critical micro-frontend for the myRedVest POS platform (Lowe's serves ~16M customer transactions a week across 1,700+ stores). Specializes in micro-frontend architecture, test automation for Electron desktop applications, API/BFF integration, and applied AI — including agentic architecture POCs for POS guided selling and autonomous test generation.

CORE SKILLS

- **Languages:** TypeScript, JavaScript (ES6+), Python, HTML5, CSS3
- **Frontend & Platform:** React, Redux / Redux-Saga, React Query, Vue.js, Angular, Webpack Module Federation (Micro-Frontends)
- **Backend & APIs:** Node.js, Backend-for-Frontend (Micronaut / Spring Boot), REST APIs, LoopBack
- **Testing & Automation:** Playwright (Electron native apps), Jest, React Testing Library, Mock Service Worker
- **AI / Agentic:** Agentic system architecture (Python, Google agent SDK), autonomous test spec generation
- **Delivery & Observability:** GCP, CI/CD (Jenkins, Azure DevOps), Webpack, Kibana / Grafana / ELK, FullStory

PROFESSIONAL EXPERIENCE

Lowe's Companies Inc. — Senior Software Engineer, Lead Developer (myRedVest Showroom POS) | Apr 2022 – Present

- Lead Developer for a team of 5 engineers (3 onshore, 2 offshore) on the Showroom POS team; single point of contact for a core area of the checkout transaction flow.
- Own the Customer Lookup Micro-Frontend — a checkout-critical capability covering customer add/edit, addresses, preferred contact method, tax-exempt and military discount, invoice/estimate recall, and adding customers to cart.
- Built the Customer MFE from scratch using Webpack Module Federation, custom events, and React Query; consumed by 20+ teams across Lowe's applications.
- Designed and built the end-to-end Playwright automation suite for the myRedVest POS desktop application (Electron), covering the checkout transaction and customer-lookup flows — catching UI regressions before they reach stores and significantly reducing production incidents.
- Designed a POC for an autonomous test spec generator to automate creation of test specifications for POS testing.
- Designed an agentic architecture for POS guided selling — guiding store associates to the next step to complete the transaction journey — using Python and the Google agent SDK (Lowe's SDK version).
- Integrated the checkout micro-frontends with common-layer BFF services (Micronaut / Spring Boot) as a consuming client — defined frontend data requirements and contracts against those APIs.
- First point of contact for production issues; change coordinator taking features through all stages to production — creating CRQs, validating deployments, and presenting in CAB meetings for approvals up to Director/CIO level.

- Contributed generic components to the myRedVest design system with the UX team; tuned application performance (Webpack/gzip configuration, FullStory, Kibana, Grafana, ELK instrumentation).

Atos Syntel Inc. — Consultant | Nov 2020 – Apr 2022

- McKesson (AIMS): Full-stack development with a Vue.js front-end focus — reusable Vue single-file components, Vuex state management, Element UI; resolved Okta Verify issues on the JavaScript side.
- Baylor Scott & White Health: React applications using React Query and Mock Service Worker; reusable React components for the bswhealth.com Sitecore migration and a .NET-based clinical trials application.

SLK America Inc. — Senior Software Engineer | Sep 2018 – Nov 2020

- First Horizon Bank: Front-end developer for the Loan Operations BPM tool — Angular 6 / TypeScript with a Node.js LoopBack API backend; built reusable Angular component libraries (forms, designer, eTime).
- M&T Bank: Digital banking products in React — e-statements, EZ-Money (P2P), Alexa banking integration, and a feature-flag dashboard; proposed the Git flow for TFS to Azure DevOps migration; implemented WCAG/ADA accessibility standards.

Valley Soft Systems Inc. — Software Engineer | Sep 2017 – Aug 2018

- Apple (Talent Point): Full-stack Meteor / React / Node.js / MongoDB work on Apple's careers recruitment platform.
- Cisco (DNA Center): React/Redux product UI and the shared Cisco-DNA-react-components library (data table, site selector, scheduler).

Alliance Information Technology Inc. — Software Engineer / Front-End Developer | Aug 2016 – Sep 2017; Jun 2014 – Aug 2015 (India)

- Symantec (Rev-Pro): React/TypeScript executive financial dashboard used for C-level operational reporting; earlier front-end development (JavaScript/jQuery) for regional vendor dashboards in India.

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- **Sealgraf —** authorization layer for AI agents: owner-set limits checked before spending, owner approval the agent cannot fake, sealed verifiable receipts · LIVE product at sealgraf.com · integration packages live-tested end-to-end on six agent frameworks (Dify, CrewAI, LangChain, n8n, Composio, Vercel AI SDK)
- **Doorway —** EAP access agent: eligibility verified, benefit explained with citations, steerable matching, approval-gated booking, member-visible privacy ledger · LIVE working demo (via saiarunglabs.com) · 44 tests
- **Sector products —** Dossier (travel recovery) · ListProof (listing verification) · JobProof (local-services evidence) · Reclaim (rewards disputes) · Refile (claims appeals) · Destorm (incident triage) · ModSeal (moderation record): working prototypes with rules enforced in code and approval gates · case studies at saiarunglabs.com · 334 tests
- **Reify —** open-source local-first memory engine for AI agents · github.com/saiarung27/reify · 442 tests
- **AgentFoundry —** agent evaluation harness · ~195 tests

EDUCATION

- M.S., Computer Information Systems & IT — University of Central Missouri, Dec 2016
- B.Tech, Mechanical Engineering — Anurag Engineering College, 2014